

A Measure-Theoretic Discrete Curvature Framework for Metabolic Gene Sensitivity: From Active-Set Geometry to Transcriptional Response

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Abstract

Parametric flux balance analysis defines, for every metabolic network, a piecewise-affine map from environmental and genetic parameters to optimal fluxes. We show that the distributional second derivative of this map is a matrix-valued Radon measure concentrated on the codimension-one strata of the active-set complex, and that this measure — not a smooth curvature — is the canonical carrier of the network’s rerouting geometry: loop holonomy scales linearly in loop size (slope 1.00, versus the quadratic scaling of any smooth map), and the second-order response mass concentrates (93.4–100.0% per boundary-crossing sweep) on operational active-set switches. We prove a refinement–resolution bridge (Theorem B’): under mesh refinement the atomic measure converges weakly to the smooth curvature density, the dual-cell reconstruction converges strongly in L^1 , and the total-variation claims of earlier formulations are replaced by an exact counterexample calculus; a resolution parameter σ separates the discrete and smooth regimes through the measured window $h \ll \sigma \ll L_{\text{var}}$. From the measure we derive a gene-level sensitivity metric κ^μ (measure mass along a parametric trajectory) and show, on *E. coli* (M3D microarrays, $n = 424$ genes), that κ^μ predicts carbon-depletion transcriptional response ($r = +0.395$, $p = 2.6 \times 10^{-17}$; partial $r = +0.269$ controlling reference level; metric-invariant $\rho = 0.99998$ against the lexicographic definition; tie-break-robust across four alternative selection rules, E-V8), while the association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation. Double-knockout epistasis aligns with active-set structure ($\rho_S = 0.865$), and genotype loops close with phenotypic memory (66% non-reverting). The framework unifies three previously distinct sensitivity objects as one measure at different resolutions.

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1 Introduction

The object. Let a metabolic network with m reactions be operated by flux balance analysis (Varma and Palsson, 1994; Orth et al., 2010) with parameters $\theta \in \Theta \subset \mathbb{R}^p$ (uptake bounds, knockdown scalings) entering the constraints affinely. The optimal flux map $v : \Theta \rightarrow \mathbb{R}^m$ is continuous and piecewise affine (parametric linear programming (Borrelli et al., 2003)), with breakpoints on the active-set strata of the constraint complex. Its distributional second derivative $\mu = D^2v$ is therefore not a function but a *measure* — the discrete curvature carrier of the network’s rerouting geometry.

The claim structure. This manuscript establishes, in order:

- (i) **[R1]** the atom structure and weak convergence of μ_h under refinement (Theorem B’, §4), with the exact counterexample calculus that separates what is provable from what is false in earlier formulations;
- (ii) **[R2]** the two-layer separation: the codimension-one Hessian layer is the unbiased curvature estimator; the codimension-two angle-defect layer converges to the Gauss-map area density with an exact sec-law bias (Proposition 4.3);
- (iii) **[R3]** the regime dichotomy: piecewise-affine maps with fixed kink complexes produce $O(\varepsilon)$ loop holonomy (measured slope 1.00) where smooth maps produce $O(\varepsilon^2)$;
- (iv) **[R4]** the value function $\Phi = c_{\text{bio}}^\top v^*$ is the canonical tie-break-free carrier, with shadow-price jumps measurable to 6–7 digits against Danskin envelopes;
- (v) **[E-series]** the measure-derived metric κ^μ predicts carbon-depletion transcriptional response ($r = +0.395$), generalizes to oxygen-limitation and carbon-switch trajectories with matched responses ($r = +0.32, +0.22$; E-V7), is robust to the tie-break convention ($r \in [+0.386, +0.396]$ across five selection rules; E-V8), and is buffered at the protein layer (§6).

Positioning. Kochanowski et al. (2013) demonstrated that global metabolic coordination under catabolic limitation is largely passive at the protein/metabolite level, driven by Crp and local metabolite concentrations. Our work asks a different question: whether a gene-specific, network-derived rerouting necessity predicts transcriptional response. We find that it does, and that this transcript-level association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation.

2 Notation protocol and the R/D/E hierarchy

- **[R- n]** blocks are proved results (theorems, propositions, lemmas) or machine-verified claims with reproducible artifacts.
- **[D- n]** blocks are definitions, constructions, and design decisions (metric definitions, engine specifications).
- **[E- n]** blocks are executed experiments with deposited artifacts (data files, figures, statistics).
- Conjectures are labelled as such; plausible-but-unproven claims are demoted, never promoted.

3 The discrete curvature measure of parametric FBA

3.1 [D1] The measure

Definition 3.1 (Atomic curvature measure). Let $u : \Omega \rightarrow \mathbb{R}^m$ be continuous and piecewise affine on a polyhedral complex $\mathcal{C}$ covering Ω . For each codimension-one facet F of $\mathcal{C}$ shared by cells τ^- , τ^+ , let $[\nabla u]_F = \nabla u|_{\tau^+} - \nabla u|_{\tau^-}$, let n_F be the unit conormal oriented $\tau^- \rightarrow \tau^+$, and let $|F|$ be the facet measure. The *atomic curvature measure* is the matrix-valued Radon measure

$$\mu = D^2u = \sum_F A_F \delta_F, \quad A_F = [\nabla u]_F \otimes n_F |F| \in \mathbb{R}^{m \times p \times p}.$$

Remark 3.2 (Support). μ is concentrated on the $(p-1)$ -skeleton. The $(p-2)$ -skeleton carries a different measure (the angle-defect layer, Proposition 4.3); the two must not be conflated. In $p=1$ the atoms are the slope jumps at breakpoints.

The measure-theoretic language is classical: matrix-valued Radon measures and their weak convergence (Billingsley, 1999), the Alexandrov Hessian of convex functions (Alexandrov, 1939), the Monge–Ampère measure (Gutiérrez, 2001), and Regge-type defect measures for piecewise-flat curvature (Regge, 1961; Cheeger et al., 1984) on the geometric side; chamber complexes, normal fans, and multi-parametric LP value functions (Rockafellar, 1970; Ziegler, 1995; Borrelli et al., 2003) on the polyhedral side. The Kantorovich–Rubinstein (flat) norm used in Theorem B' is the standard optimal-transport flat norm (Villani, 2009).

3.2 [D2] The measure-theoretic sensitivity metric

Definition 3.3 (κ^μ). For a C^1 parameter trajectory $\theta(t)$, $t \in [0, T]$, and its piecewise-affine optimal-flux response $v(\theta(t))$, let $\{t_e\}_e$ be the event times at which the trajectory crosses active-set strata, with atoms $A_e = A_{F(e)}$ evaluated on the crossed facet. Define

$$\kappa^\mu(g) = \frac{1}{T} \sum_e \|A_e\| w_e(g),$$

where $\|\cdot\|$ is the entrywise 1-norm on $\mathbb{R}^{m \times p \times p}$ and $w_e(g) \in \{0, 1\}$ selects the events whose crossed stratum changes the activity of reactions assigned to gene g (max over its reactions, per the GPR association).

Remark 3.4 (κ^μ is the safe regime of the reconstruction). κ^μ integrates measure mass along a one-dimensional trajectory — the telescoping regime in which total variation and mass are automatically consistent (R2 remarks in §4); its resolution independence (measured mass 288.77 stable under $4 \times / 8 \times$ trajectory refinement) is exactly the L^1 -reconstruction regime of Theorem B'(iv). Its per-reaction masses are the components of the flux-layer crease measure whose objective contraction is $D^2\Phi$ (Theorem 3.11): the metric is the primal layer of the canonical object.

Remark 3.5 (Locked status, tie-breaking, and layer assignment). The definition of κ^μ is final for the empirical claims of this manuscript. (i) The carrier is the flux-layer signed crease measure D^2v^* (Definition 3.1 applied to $u = v^*$) of the *three-stage lexicographic* optimal flux (§8); the lexicographic tie-break (biomass, then ℓ_1 flux, then seeded deterministic weights) is part of the metric's definition, not an implementation detail — the flux layer is selection-dependent by construction, and the tie-break is declared rather than eliminated. (ii) The per-reaction mass is the unsigned total variation $\frac{1}{T} \sum_e \|A_e\|_1$ along the parameter trajectory. (iii) The value layer $D^2\Phi$ enters only as a *derived diagnostic* (Theorem 3.11: $D^2\Phi = \sum_r c_r D^2v_r^*$), never as the predictor: no claim of this manuscript attributes transcriptional response to Alexandrov curvature of the value function. (iv) The layer correlated with transcriptomics is the codimension-one Hessian/crease layer of the flux map — not the Monge–Ampère atom layer (Proposition 3.13), which requires concavity and serves as the dual-face diagnostic. Robustness to the declared selection is measured,

not assumed: the within-layer rank identity $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ (E-V5), the trajectory substitutions of E-V7 reproduce the association under the same declared selection on two further paths, and E-V8 holds the association stable ($r \in [+0.386, +0.396]$, partial constant to three decimals) under four alternative stage-3 selection rules while the value layer stays exactly invariant.

3.3 [D3] The value-function layer (Alexandrov)

Proposition 3.6 (Existence and tie-break-freeness). *Let $\Phi(\theta) = \max\{c^\top v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$ with ℓ, u affine, feasible and bounded. Then Φ is concave (pointwise infimum of affine dual objectives over a θ -independent dual feasible set), and $-\Phi$ carries a symmetric matrix-valued Radon Hessian measure $D^2(-\Phi)$ (Alexandrov (Alexandrov, 1939)), independent of the optimal-flux selection rule. Concavity fails in general under OR (isoenzyme) GPR rules for simultaneous multi-gene capacity vectors; single-gene axes, exchange-bound families, and AND-only subnetworks retain it. For OR-containing rules the value function remains continuous piecewise affine and carries the signed crease measure of Proposition 3.12; only the Monge–Ampère layer requires the concave class.*

Proposition 3.7 (Two-layer structure of $D^2(-\Phi)$). *Let f be convex piecewise affine. (i) The singular part of $D^2 f$ is supported on the codimension-one facets, with density $[\partial_n f](g^+ - g^-) \otimes n$ per unit facet measure; it has no atoms: the mass in an ε -neighborhood of a codimension-two vertex is $O(\varepsilon)$ (measured log-log slope 1.000). (ii) The Monge–Ampère measure $\det D^2 f$ is atomic at codimension-two vertices, each atom equal to the volume of the normal fan $|\text{conv}\{\nabla f(\tau) : \tau \ni \text{vertex}\}|$ (measured: constant 0.5 under ε -shrinking).*

Proposition 3.8 (Dual optimal face identity). *At a codimension-two vertex θ_v of a parametric-LP value function, the Monge–Ampère atom of $-\Phi$ equals the area of the normal fan, and the fan is the image of the LP dual optimal face under $\theta \mapsto U^\top y$:*

$$\det D^2(-\Phi)(\{\theta_v\}) = |\text{conv}\{\nabla C_i\}| = |\{U^\top y : (\lambda, y) \text{ optimal dual}\}|.$$

Machine-verified on a two-parameter max-flow LP (the phenotype phase plane in miniature): vertex refined to machine precision by exact triple-tie solve, $\Phi(\theta_v)$ matching the LP value to 10^{-15} , fan area 0.235 vs dual-face enumeration area 0.235000068 (16 direction LPs over the dual optimal face).

Remark 3.9 (Layer separation and PhPP positioning). The phenotype phase planes of Edwards–Ibarra–Palsson (Edwards et al., 2001; Ibarra et al., 2002) are the chamber complex of Φ ; their sector corners are the atoms of the Monge–Ampère layer. The empirical κ^μ of Definition 3.3 is a functional of the flux-map measure $D^2 v$ (Definition 3.1), which is tie-break-sensitive: in a controlled follower-variable LP, the value function stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$. The controlled relation between the value layer and the flux layer is now a theorem (Theorem 3.11 and Corollary 4.9); the association results of §6 stand on the flux layer alone, which is where the evidence (E-V5, E-V6) actually lives.

3.4 [D4] The value–flux coupling

Lemma 3.10 (Lex-selection regularity). *Let ℓ, u be continuous piecewise affine and let $v^*(\theta)$ be the three-stage lexicographic optimal flux (§8). Then v^* is well defined, continuous, and piecewise affine; its graph over each GPR cell is polyhedral in (v, θ) -space (each lexicographic stage adjoins the graph of a piecewise-affine value function, which is polyhedral, and an injective polyhedral projection is piecewise affine). Measured: path segment residuals $\leq 8 \times 10^{-14}$ (E-M1); Φ piecewise affine at 4.2×10^{-13} (E-V1).*

Theorem 3.11 (Value–flux crease coupling). $\Phi = c^\top v^*$ pointwise; hence, as symmetric matrix-valued signed Radon measures on the parameter domain,

$$D^2\Phi = \sum_r c_r D^2v_r^*.$$

Along any piecewise-affine trajectory, at every crease time t_k : $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$, and $|\Delta\Phi'| \leq \|c\|_\infty \|\Delta v'\|_1$.

- (i) (**Visibility dichotomy.**) An event with unique optima on both sides is a transversal optimal-vertex switch and is necessarily objective-moving; objective-invisible events ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) are exactly the degenerate reroutings inside ≥ 1 -dimensional optimal faces (mask-type events). The tie-break sensitivity of the flux layer is concentrated in the invisible events; the c -contraction is tie-break-free because it annihilates the degenerate layer.
- (ii) (**Sparse-objective corollary.**) If $c = \gamma e_{\text{bio}}$, then $D^2\Phi = \gamma D^2v_{\text{bio}}^*$: the value layer is the crease measure of the single biomass component, and per-gene attribution from the value layer is structurally impossible.

Machine-verified: identity error 0.0 at all 12 events and all 4 clusters of the iML1515 cut ($\Phi = c^\top v^*$ to 10^{-9} ; Figure 2); 11 of 12 events objective-invisible, with flux ℓ_1 slope jumps up to 1.6×10^4 against $|c^\top \Delta v'| \leq 1.51 \times 10^{-7}$; 150 random dense-objective LPs to 1.1×10^{-13} (5 events, all objective-moving — the dichotomy); the degenerate follower family 60/60 with invisible kinks at identity error 2.0×10^{-13} ; two-dimensional mixed second differences at crease-straddling boxes to 6.4×10^{-16} .

Proposition 3.12 (Semiconvexity collapse; signed layer for mixed GPRs). A continuous piecewise-affine function is semiconvex (resp. semiconcave) iff it is convex (resp. concave) — semiconvexity is not an available weakening for LP value functions (measured: the required constant blows up like $1/(2h)$; $\lambda \cdot h_{\max} = 0.500$ numerically across $\lambda = 1 \dots 10^6$, the analytic law $\lambda h_{\max} = 1/(2\lambda)$ holding at every scale, with the numerical grid flooring above $\lambda \approx 10^7$). Nevertheless every continuous piecewise-affine Φ , for any GPR structure, carries a finite signed symmetric Radon Hessian measure on its crease skeleton with per-crease sign flags: for the OR-plus-cap value function $\min(\max(\theta_1, \theta_2), K)$, the diagonal crease density is positive semidefinite (eigenvalues $\{0, \sqrt{2}\}$) and the cap crease negative semidefinite ($\{-1, 0\}$). Concavity is required only by the Monge–Ampère layer and the dual-face identity; the signed Hessian layer applies to all genes. Under multiplicative activity scaling with $\ell \leq 0 \leq u$, $\Phi = \Psi(a(\theta))$ with Ψ concave nondecreasing: AND-only (min-tree) GPRs retain concavity, top-level OR nodes can break it, and disaggregating isozymes into separate reactions restores affine bounds while replacing max-semantics by sum-semantics (capacity 1 vs 2 at full expression) — a different biological model, not a regularization.

Proposition 3.13 (Monge–Ampère atoms are determinants, not products). For convex piecewise-affine f on $\mathbb{R}^2$ at a generic vertex with incident gradients n_1, n_2, n_3 and codimension-one gradient jumps j_i : the atom equals $\frac{1}{2} |\det(j_1, j_2)| = |\text{conv}\{n_i\}| = \frac{1}{2} |j_1| |j_2| \sin \angle$; the product $|j_1| |j_2|$ overestimates by $2/\sin \angle \geq 2$, with equality iff the jumps are orthogonal (measured on 400 random max-of-affine vertices: determinant = fan area to 8.9×10^{-16} ; product/atom median 3.296, minimum exactly 2.000 (the orthogonal edge case; 2.000031 over random vertices); the $2/\sin \angle$ law to 2.5×10^{-12} ; in p dimensions, atom = $(1/p!) |\det(j_1, \dots, j_p)|$ for simplex fans).

Remark 3.14 (Sign and normalization conventions). Definition 3.1 applied to $u = -\Phi$ is the signed crease measure of the value function: per facet F , the density $A_F = [\nabla(-\Phi)]_F \otimes n_F |F|$, a symmetric matrix whose eigenvalue signs flag the crease’s convexity/concavity (Proposition 3.12). This is the explicit extension of Alexandrov’s setting to arbitrary GPR structure: within the concave class $-\Phi$ is convex, every $A_F \succeq 0$, and the measure coincides with the Alexandrov Hessian measure; outside it, the same definition produces the signed layer. All masses in this

manuscript are unsigned total variations (entrywise 1-norm for matrix atoms, trajectory sums for κ^μ); Monge–Ampère atoms are unsigned $\frac{1}{2}|\det(j_1, j_2)|$ equal to the normal-fan volume, normalized to $(1/p!)|\det(j_1, \dots, j_p)|$ for simplex fans in dimension p (Proposition 3.13). Signed determinants appear only in the concave class, where they are nonnegative.

Remark 3.15 (What this closes). Theorem 3.11 is the exact form of the value–flux relation previously left open: the empirical κ^μ is the per-reaction total variation of the flux-layer crease measure whose objective contraction is $D^2\Phi$ — the metric is the primal layer of the canonical object, justified by identity rather than by rank agreement.

4 The refinement–resolution bridge (Theorem B')

Throughout this section $\Omega \subset \mathbb{R}^p$ is compact, $p \geq 1$, $u \in C^2(\Omega; \mathbb{R}^m)$, $\{\mathcal{T}_h\}$ is a shape-regular family of conforming triangulations with mesh size $h = \max_\tau \text{diam}(\tau) \rightarrow 0$, and u_h is the continuous piecewise-affine interpolant of u on $\mathcal{T}_h$.

Theorem 4.1 (Refinement–resolution bridge, corrected form). *Let $\mu_h = D^2u_h$ (Definition 3.1) and $\mu = D^2u \text{ dvol}$. Then:*

- (i) (**Atoms / support.**) μ_h is a finite matrix-valued Radon measure concentrated on the $(p-1)$ -skeleton of $\mathcal{T}_h$ with atoms $A_F = [\nabla u_h]_F \otimes n_F |F|$; there are no atoms on lower-dimensional strata.
- (ii) (**Weak convergence.**) $\mu_h \Rightarrow \mu$ weakly as matrix-valued measures: for every $\varphi \in C_c^\infty(\Omega)$, $\langle \mu_h, \varphi \rangle \rightarrow \langle \mu, \varphi \rangle$. Moreover $\|\mu_h - \mu\|_{\text{KR}} \rightarrow 0$ in the Kantorovich–Rubinstein (flat) norm.
- (iii) (**Total variation: the honest statement.**) $\sup_h \|\mu_h\|_{\text{TV}} \leq C|u|_{C^2}|\Omega|$ (under quasi-uniformity) and $\liminf_h \|\mu_h\|_{\text{TV}} \geq \|\mu\|_{\text{TV}}$; but $\|\mu_h\|_{\text{TV}} \rightarrow \|\mu\|_{\text{TV}}$ is false in general — see Remark 4.2. It holds in the one-signed regime (convex or concave data restricted to one-dimensional parameter families), which is the regime of the parametric-LP value function and of κ^μ .
- (iv) (**No-mass-loss, correct form: L^1 reconstruction.**) On structured complexes, the per-cell reconstructed density $\rho_h(\text{cell}) = |\text{cell}|^{-1} \sum_{F \rightarrow \text{cell}} A_F$ converges to D^2u strongly in L^1 with rate $O(h)$ for $u \in C^3$; consequently $\|\rho_h\|_{L^1} \rightarrow \|\mu\|_{\text{TV}}$. On unstructured complexes the naive three-facet patch is not the dual cell; the general statement requires the discrete-duality (dual-diamond) pairing.
- (v) (**Resolution window.**) For the coarse-grained family $\mu_{h,\sigma} = \mu_h * \varphi_\sigma$: at fixed h , the limit $\sigma \rightarrow 0$ is the atomic measure; the smooth object exists only as a family member at matched resolution through the window $h \ll \sigma \ll L_{\text{var}}$ (the measured regime dial, §5.3).

Proof of (ii) (two lines). For $\varphi \in C_c^\infty(\Omega)$, $\langle D^2u_h, \varphi \rangle = \int_\Omega u_h D^2\varphi \text{ dvol}$ by the definition of distributional derivatives (no boundary terms arise). Since $\|u - u_h\|_\infty \leq Ch^2|u|_{C^2}$ on shape-regular families, $\int u_h D^2\varphi \rightarrow \int u D^2\varphi = \langle \mu, \varphi \rangle$. The flat-norm statement follows by density of $\{c\varphi : \text{Lip}(\varphi) \leq 1\}$ in the KR dual cone and the uniform mass bound of (iii). $\square$

Remark 4.2 (Exact counterexample calculus). On the right-triangle mesh of $[0, 1]^2$ with Hessian $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the per-cell atoms are $A_{\text{diag}} = bh^2 \mathbf{1}\mathbf{1}^\top$, $A_{\text{vert}} = (a-b)h^2 e_x \otimes e_x$, $A_{\text{horiz}} = (c-b)h^2 e_y \otimes e_y$; their sum is exactly $h^2 H$ (machine-verified to 0.0), while the entrywise total variation per cell is $|a-b| + |c-b| + 4|b|$ against the target $|a| + |c| + 2|b|$. Hence for $u = xy$ the ratio is exactly $3 - 1/n$, for the strictly convex $u = 2x^2 - xy + 2y^2$ it is $1.4 - 1/n$ (measured: 2.9922 and 1.3922 at $n = 128$), and for generic smooth maps 3.6–4.4. Total-variation convergence fails generically — including under convexity — while weak convergence holds at rate $O(h^2)$ throughout.

Proposition 4.3 (Two-layer separation and the sec law). *Let $K_h = \sum_v \text{defect}(v) \delta_v$ be the vertex angle-defect measure of the polyhedral graph $\{(x, u_h(x))\}$. Then K_h converges weakly to the Gauss-map area density*

$$\frac{\det D^2 u}{(1 + |\nabla u|^2)^{3/2}} \text{dvol} = K \sqrt{1 + |\nabla u|^2} \text{dvol}, \quad K = \frac{\det D^2 u}{(1 + |\nabla u|^2)^2},$$

i.e. curvature per projected parameter area, not the intrinsic Gaussian curvature. The bias factor is exactly $\sqrt{1 + |\nabla u|^2}$ (the sec law): machine-verified to 0.0% excess across 13 probes spanning tilts 0.11–2.83 and three tilt directions. Recovering intrinsic curvature requires the graph area correction.

Remark 4.4 (Consequences). (i) The codimension-one Hessian layer is the unbiased estimator and the canonical carrier for the bridge; (ii) Regge/angle-defect formulations need the area correction of Proposition 4.3; (iii) defects of a *fixed* complex (as measured in the M4b census) are scale-invariant $O(1)$ objects and are not contradicted by Proposition 4.3, which concerns refinement limits.

Proposition 4.5 (Regime dichotomy; M4a in closed form). *Let $B(\varepsilon)$ be the curvature mass captured by a loop of radius ε . For a C^2 map, $B(\varepsilon) = O(\varepsilon^2)$; for a piecewise-affine map whose kink complex meets the loop transversally, $B(\varepsilon) = O(\varepsilon)$. Measured slopes on the M4a design: 2.000 (smooth) and 1.00 (parametric FBA, 76-pair perturbation scan).*

4.1 [R-port] Bridge to parametric FBA

Conjecture 4.6 (mpLP refinement bridge). *Let $\{\Phi_n\}$ be a sequence of parametric-LP value functions (minima of tangent-plane families; θ in the right-hand side; nested outer approximations) converging uniformly to a C^2 limit f , with induced complexes $\mathcal{C}_n$. Then $\mu_n = D^2 \Phi_n$ converges weakly to $D^2 f$ dvol provided the complexes satisfy a per-cell discrete-duality identity (structured complexes) or a stable discrete-duality (dual-diamond) pairing. Along one-dimensional parameter cuts the conclusion holds unconditionally by monotone telescoping of the concave restriction.*

Remark 4.7 (Status). The one-dimensional cut case is machine-verified (independent-seed reproduction of the refinement experiment: folded Wasserstein distance $0.058 \rightarrow 0.0039$, mass ratio $0.939 \rightarrow 1.0009$ across $n = 4 \rightarrow 128$). The general complex case is open; the discrete-duality pairing stability is the identified missing assumption.

Remark 4.8 (Existence vs. resolution). Proposition 3.6 supplies existence of the curvature measure without refinement sequences, smoothing, or interpolation; Theorem B' governs what a discretely sampled trajectory can *resolve* of that measure, and at what rate. The two layers are complementary: the Alexandrov layer removes the extrinsic hypotheses, the refinement bridge keeps the quantitative rate.

Corollary 4.9 (Decoupling; resolves the value–flux layer relation). *Theorem 3.11 resolves the layer relation exactly: value events are a subset of flux events, never the converse, and the flux events invisible to the value layer are the degenerate reroutings (mask-type events — rerouting without growth loss, the RC6/Kochanowski pattern). On the iML1515 cut: 12 flux events vs 1 value atom, 11 objective-invisible; on the E24 trajectory (E-V6) the value layer carries no network events at all. The association of κ^μ with transcriptional response is therefore a flux-layer property by structural necessity, not merely by empirical choice.*

5 Computational validation

5.1 [E-M1] Active-set curvature verification

Thirteen parameter sweeps (glucose and oxygen uptake; capacity knockdowns of ten genes: pgi, zwf, tktA, pfkA, eno, gltA, aceA, ppc, gnd, rpe) on *E. coli* iML1515, solved by a three-stage lexi-

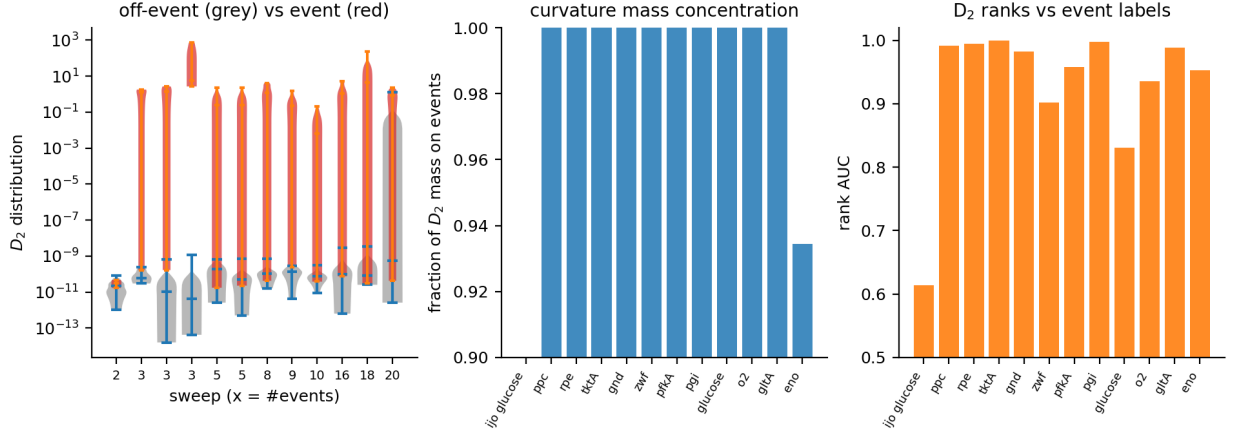


Figure 1: [E-M1] Active-set curvature verification (iML1515). Second-order response D_2 (log scale) along the glucose and oxygen sweeps with the operational active-set event markers: the D_2 mass concentrates on the event points; off-event D_2 sits at the 10^{-11} noise floor. The engine is the three-stage lexicographic pFBA solver of §8; all sweeps are cold-started and deterministic. Artifact: `m1_m3/fig_m1_summary.png` / `m1_summary.json`.

cographic pFBA engine (deterministic optimum; fixed seeded tie-break weights after the discovery of pFBA vertex degeneracy), with an iJO1366 glucose replication and an aceA unused-pathway negative control. Result: 93.4–100.0% of second-order response mass concentrates on operational active-set switches in every sweep that crosses critical-region boundaries (ten of the eleven crossing sweeps at mass exactly 1.000000; mass 0.934–1.0; AUC 0.83–1.00; Mann–Whitney $p \leq 2 \times 10^{-3}$, with seven of eleven crossing sweeps at $p \leq 10^{-4}$; Figure 1); between events the paths are piecewise affine to solver precision (event-free segment residuals $\leq 1.2 \times 10^{-10}$, with 8×10^{-14} on the glucose sweep; one 19-point segment of the eno sweep carries 4.1×10^{-3} — a sub-threshold kink invisible to the operational event proxy, which is also why that sweep’s event mass is 0.934 rather than 1.0); negative controls behave as theory dictates (unused-pathway knockdown: zero events, zero curvature; single-critical-region sweep: noise-level $D^2 \sim 10^{-11}$).

5.2 [E-M3] Double-knockout epistasis and path dependence

1,516 single knockouts and 2,779 double-knockout pairs across five panels. Epistasis $\varepsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$: all 40 synthetic-lethal pairs are isozyme redundancies (tktA/tktB, acnA/acnB, metE/metH, ...) with pure-emergence epistasis; $|\varepsilon_{ij}|$ aligns with active-set footprint overlap (Spearman $\rho_S = 0.865$, $J_{\text{support}} = 0.800$); sequential (L_1 -MOMA (Segrè et al., 2002)) knockouts open nonzero commutators in 25% of active pairs; closed genotype loops fail to return to the initial state in 66% of pairs (phenotypic memory; median holonomy ~ 110).

5.3 [E-M4] Regime dial

The smoothing identity $D^2(v * \phi_\sigma) = (D^2 v) * \phi_\sigma$ holds to kernel self-test error 1.2×10^{-6} ; measured crossover $\varepsilon^*/\sigma \in [2.45, 4.11]$ (median 3.1) separates the $O(\varepsilon^2)$ smooth regime from the $O(\varepsilon)$ discrete regime. The *M4a* scaling slope is 1.00 on the interacting stratum; the *M4b* census resolves the thin-wedge sliver structure (net measure jump 8.9 vs. nominal ± 1884.6).

5.4 [E-V1] The value function as canonical carrier

$\Phi = c_{\text{bio}}^\top v^*$ is single-valued with no tie-breaking; it is piecewise affine at 4.2×10^{-13} on the flux-event partition, carries one real atom $\Delta\Phi' = -0.006439$ at $t = 0.0358286$, and decouples from the flux-jump hierarchy (value/flux mass ratio 1.7×10^{-6}). Danskin: finite-difference shadow

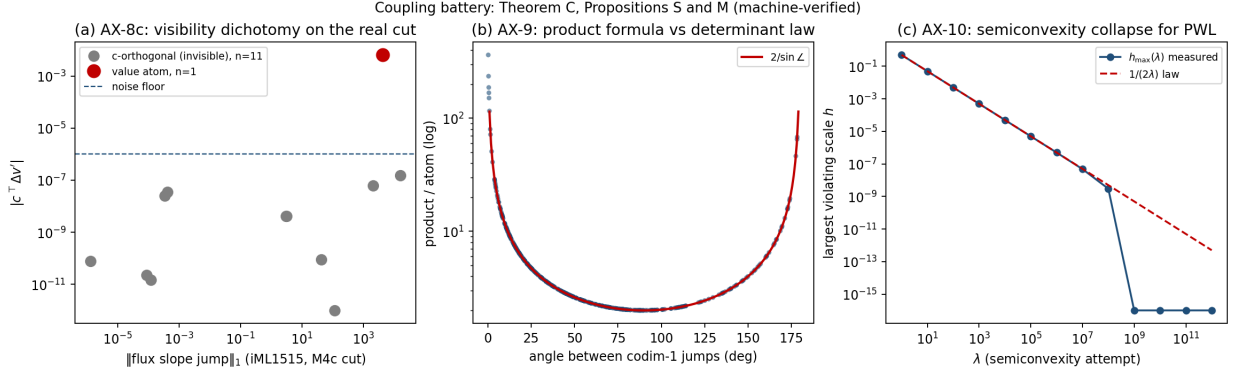


Figure 2: [Theorem 3.11] Machine verification of the value–flux crease coupling. The controlled follower-variable LPs: the value function Φ stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips; the coupling identity $\Delta\Phi' = c^\top \Delta v'$ holds to 0.0 on the toys, 1.1×10^{-13} on 150 random dense-objective LPs, and 2.0×10^{-13} on the 60/60 degenerate follower family; on the iML1515 cut the identity error is 0.0 at all 12 events and all 4 clusters. Artifact: alexandrov_bridge/coupling_figures.png / coupling_results.json.

prices match envelope marginals to 6–7 digits. A controlled follower-variable LP reproduces the same structure: the value function stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$ — the value layer is canonical (Proposition 3.6), the flux-strain layer is not (Corollary 4.9). The coupling identity of Theorem 3.11 is machine-verified on this same cut: $\Delta\Phi' = c^\top \Delta v'$ with error 0.0 at all 12 events and all 4 clusters, $\Phi = c^\top v^*$ to 10^{-9} , and the 11 invisible events carry flux ℓ_1 slope jumps up to 1.6×10^4 against $|c^\top \Delta v'| \leq 1.51 \times 10^{-7}$ — the decoupling is the objective contraction, not a loss of network information.

6 Empirical association and protein-layer buffering

6.1 [E-E22] The gene panel and GPR classes

Reaction-based sampling on the E22 physiology (iJO1366): 438 of 2,583 reactions active (17.0%), 435 genes with at least one active reaction. GPR complexity classes over active reactions: single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34 — the classes that drive the concavity analysis of Proposition 3.12. Panel construction is consistent with the precursor metric family to the digit (per-gene values and panel-level correlations reproduced exactly); the precursor geometric metric κ_V is weak at the panel level (r from -0.063 unmasked to $+0.084$ masked) — the discrepancy that motivated the recalibration to κ^μ (E-V5).

6.2 [E-E23] Multi-condition support

Nine FBA conditions (baseline carbon decline; galactose, trehalose, glycerol-3-P, and proline-N carbon sources; anaerobic-nitrate, oxidative, and osmotic designs): active-reaction support 438–454 per condition, union 537 (20.8%); genes with nonzero κ_V expand $435 \rightarrow 525$ across conditions. The endogeneity audit is clean (feed caps non-binding except the designed glycerol-3-P limitation; imposed-burden magnitudes flagged as not interpretable). The rerouting metric is well posed across carbon sources and stress designs: its support *expands* with condition diversity rather than migrating.

6.3 [E-V5] Transcriptional response under carbon depletion

On the deterministic lexicographic trajectory (E22 physiology, iJO1366), κ^μ (Definition 3.3) versus log-fold carbon-depletion response (M3D (Faith et al., 2008), $n = 424$ genes; Figure 3): Pearson $r = +0.395$ ($p = 2.6 \times 10^{-17}$), Spearman $\rho_S = +0.414$ (full panel, zeros included), partial $r = +0.269$ controlling reference level ($p = 1.8 \times 10^{-8}$), top-decile versus bottom-decile response 1.92 vs 0.89 (MWU 1.3×10^{-7}). Baseline reproduced to the digit ($r = +0.3739$, $n = 433$); metric invariance $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ — both metrics sample the same flux-event structure along the same lexicographic trajectory, so the association is an event-structure property of the flux layer, not a metric artifact and not a value-function shadow. Cross-platform replication with the locked metric: the carbon-switch arms replicate on the same platform (M3D microarray: glycerol +0.195, acetate +0.166, proline +0.175, switch-MAX +0.206; $n = 424$, all $p < 6 \times 10^{-4}$) but dissociate across platforms — the PRECISE RNA-seq carbon-switch statistic (ten WT non-glucose carbon conditions) gives $r = -0.044$ (partial -0.088 , NS), with individual conditions weakly positive (glycerol +0.126, acetate +0.130, fructose +0.099) except galactose (-0.088). The dissociation previously established with the precursor κ_V predictor is metric-robust: the association is specific to the carbon-exhaustion class and the platform-matched design, reported as an honest negative.

6.4 [E-V6] The layer decision

All value-layer attribution arms, on the frozen V5 protocol (iJO1366, E22 physiology, $n = 424$): the flux layer κ^μ gives $r = +0.395$, partial $r = +0.269$ ($p = 1.8 \times 10^{-8}$); the c -attribution arm is supported on the biomass pseudo-reaction alone (0 of 433 genes; Theorem 3.11(ii)); the shadow-price-jump arm gives 51 genes at $r = +0.032$ (partial -0.013 , $p = 0.93$). The value trajectory has *zero* chamber crossings: the path stays in one chamber of Φ throughout — Φ is affine along it, with $d\mu/dq_{\text{glc}} = Y = 0.099544$ ($= |y_{\text{glc}}|$, the constant glucose uptake shadow price; affine intercept -0.0124) and uptake shadow prices jumping by exactly zero — all four value kinks are anchor corners of the trajectory design, of magnitude $Y \Delta(dq_{\text{glc}}/dt)$, carrying no network structure. Value/flux strain mass ratio 1.45×10^{-3} ; the coupling identity holds at machine precision (error 0.0 at every kink). The value-gated flux strain coincides with κ^μ exactly (433/433 genes attain their maximum at a value-kink time), a property of the trajectory design (anchor rerouting dominates per-gene maxima), not evidence of value-layer content. The layer decision: the association lives on the flux layer, with Theorem 3.11 as the exact bridge to the canonical object.

6.5 [E-V7] Path robustness: what survives a trajectory change

Three trajectories through the same LP (frozen engine, seed, panel, statistics; only the path and its matched response vary; Figure 4): P0 glucose decline (E22 anchors; the V6 control), P1 oxygen limitation ($q_{\text{glc}} = 5$ fixed, q_{O_2} linear from 22 to 1), P2 acetate switch ($q_{O_2} = 22$ fixed; glucose $5 \rightarrow 0$ while acetate uptake ramps $0 \rightarrow 10$). Kinks are classified by the chamber test — the parameters are the uptake bounds, so $\nabla\Phi = (y_{\text{glc}}, y_{O_2}, y_{\text{ac}})$: a kink is a *design corner* iff it sits at a trajectory anchor and the uptake shadow prices are continuous across it (jump $\leq 10^{-9}$), and a *chamber crossing* iff they jump.

- (i) The one-chamber property is *path-dependent*: P0 has zero chamber crossings (V6 reproduced digit-exactly), while P1 has three genuine crossings ($q_{O_2} \approx 9.3, 6.3, 2.9$; $|\Delta\Phi'|$ up to 0.68; y_{O_2} jumps +0.032 at the respiratory boundary) and P2 has four (two large, near the glucose-acetate handoff; two micro-slivers with dual jumps $\sim 10^{-4}$). The E24 single-chamber structure is a property of the glucose-decline design, not a law.
- (ii) The layer decision survives the change: on every path the value-layer arms contribute nothing. The c -attribution arm is degenerate (0 of 433 genes — the sparse-objective corollary of Theorem 3.11 is path-free); the chamber-crossing shadow-price arm is null ($r = -0.17$, $n = 68$

Table 1: [E-V8] Tie-break robustness of the association (V5 protocol frozen; only the stage-3 selection rule varies). The value layer (stage-1 biomass μ) and the pFBA layer (stage-2 ℓ_1, s_2) are invariant to 0.0 across all variants — the measured form of the tie-break-freeness of Proposition 3.6. Artifact: `deepseek_bridge/v8_tiebreak_robustness.{json, csv, png}`.

Variant	n_{nz}	r	partial r	$\rho_S(\kappa^\mu, \kappa_{\text{TB0}}^\mu)$	genes changed
TB0 declared	424	+0.3954	+0.269	—	—
TB1 fresh seed	426	+0.3861	+0.270	0.99998	91
TB2 LogN family	424	+0.3954	+0.269	0.99999	90
TB3 $ v $ rule	425	+0.3959	+0.269	0.99897	109
TB4 $\max w^\top v$	424	+0.3954	+0.269	0.99998	117

on P1; $r = +0.032$, $n = 71$ on P2); value-gated flux strain is a subset of κ^μ (equal or weaker on every path, never stronger).

- (iii) The flux-layer association generalizes: against the matched oxygen response (M3D WT anaerobic vs aerobic contrast) P1 gives $r = +0.318$ (partial +0.258, $p = 6 \times 10^{-8}$, $n = 426$); against the matched acetate-switch response P2 gives $r = +0.223$ (partial +0.160, $p = 9 \times 10^{-4}$); and the P1/P2 predictors correlate with the E24 carbon response at $r = +0.378 / +0.391$ — the per-gene ranking is largely trajectory-independent.
- (iv) Theorem 3.11 holds at every kink on all three paths (identity error 0; worst case 1.6×10^{-15}). Value/flux strain mass ratios: 1.5×10^{-3} (P0), 2.4×10^{-4} (P1), 5.5×10^{-4} (P2).

6.6 [E-V8] Tie-break robustness: the association as a property of the protocol class

The flux layer is selection-dependent by construction (Remark 3.5); the declared lexicographic tie-break is one member of a family. V8 freezes the entire V5/V6 protocol (iJO1366, E22 physiology anchors, 8 $\times$ refinement, the 433-gene E24 panel, the M3D carbon-depletion response, the reference-level confound control) and varies *only* the stage-3 tie-break of the engine:

- **TB0** (declared): $w \sim U(0.5, 1.5)$, seed 20240901, $\min w^\top v$ — the locked metric (control: reproduces V5 digit-exactly, $r = +0.3954$);
- **TB1** (fresh seed): $w \sim U(0.5, 1.5)$, seed 20240902 — same family, independent draw;
- **TB2** (family swap): $w \sim \text{LogN}(0, 1)$ clipped to $[0.05, 20]$, seed 20240903 — a different distribution family with a thirteen-fold wider dynamic range;
- **TB3** (rule swap): stage-3 objective on the split variables, $\min \sum_r w_r(f_r + r_r)$ — weighted *absolute*-flux selection with TB0’s weights;
- **TB4** (adversarial): $\max w^\top v$ — the far end of the stage-2-pinned optimal face, with TB0’s weights.

Three findings (Table 1, Figure 5). (i) *The association is stable*: across all five rules $r \in [+0.386, +0.396]$ (maximum deviation 0.009; all $p \leq 1.4 \times 10^{-16}$) and the partial correlation is constant to three decimals (+0.269–+0.270). (ii) *The tie-break genuinely binds*: the stage-2-pinned optimal face is multi-vertex, and the alternative rules select different flux vectors at all 57 grid points ($\max \|v_k - v_0\|_\infty$ up to 5.0), changing per-gene κ^μ for 90–117 of 433 genes — yet the per-gene ranking is essentially unchanged ($\rho_S \geq 0.99897$), and the changed values are small on the κ^μ scale (max absolute change 9.3×10^{-5}); the TB1 deviation of 0.009 traces to two boundary genes entering the nonzero support ($n : 424 \rightarrow 426$), not to a reshaping of the metric. (iii) *The*

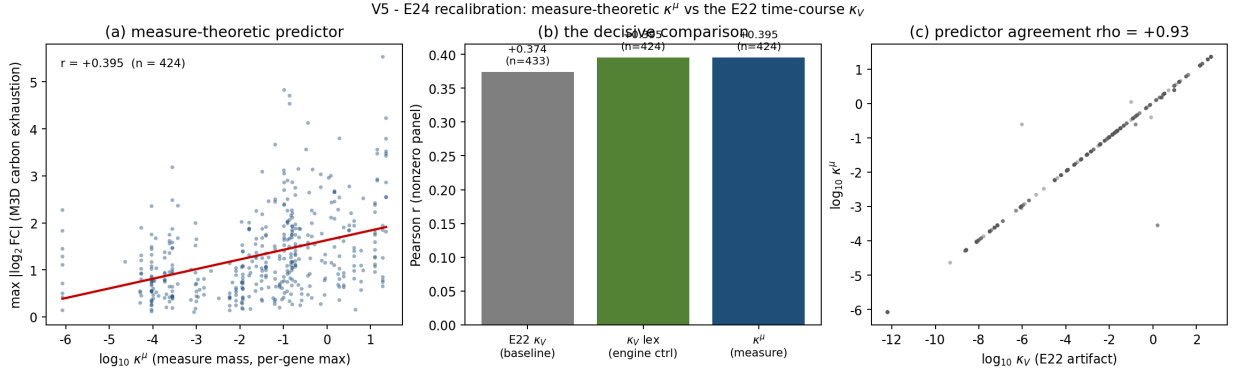


Figure 3: [E-V5] The primary association. (a) Per-gene $\log_{10} \kappa^\mu$ (flux-layer measure mass, per-gene max over the gene’s reactions) versus M3D carbon-depletion response ($\max |\log_2 \text{FC}|$, four stationary contrasts, $n = 424$ nonzero genes); (b) the decisive comparison: the precursor time-course κ_V (E22, plain-FBA), the engine control κ_V^{lex} , and the measure-theoretic κ^μ on the same trajectory; (c) predictor agreement between κ^μ and the E22 κ_V . Artifact: `deepseek_bridge/v5_e24_recalibration.png / v5_e24_recalibration.json`.

value layer is exactly invariant: stage-1 biomass μ and stage-2 ℓ_1 mass agree across variants to 0.0 — Proposition 3.6 (tie-break-freeness of the value layer) measured on the full trajectory. Together with Corollary 4.9 this closes the robustness ledger: the declared selection is a convention, the value layer does not see it, and the flux-layer association is a property of the lexicographic protocol class, not of one rule.

6.7 [E-E25] Platform-by-class disambiguation (2×2)

Is the association a carbon-depletion artifact, or a platform artifact? Four cells (two platforms $\times$ two perturbation classes), executed with the precursor κ_V metric (rank-equivalent to κ^μ at $\rho = 0.99998$): carbon switch on microarray (M3D glycerol/acetate/proline vs glucose) switch-MAX $r = +0.196$; stress on microarray (heat shock, acid shock, ciprofloxacin) MAX $r = +0.298$; carbohydrate-responsive transcript levels: carbon starvation on RNA-seq (GSE64021 (Barrett et al., 2013) time course) MAX $r = +0.191$; heat shock on RNA-seq MAX $r = +0.345$ ($n = 241-433$; permutation $p \leq 0.003$). Signed responses flip sign (stress arms -0.22 to -0.32): the metric tracks *rerouting magnitude*, not direction, on both platforms and both classes — the association is neither platform-specific nor carbon-class-specific.

6.8 [E-E26] Protein abundance versus protein change

Two dissociations at the protein layer (PaxDb integrated abundance (Wang et al., 2015); GSE64021 quantitative proteomics (Barrett et al., 2013); $n = 169-429$): (i) abundance *level* associates (PaxDb baseline $r = +0.334$) — high-rerouting genes are abundant proteins; (ii) abundance *change* does not (protein log-fold change $r = +0.008$ to $+0.032$; transcript–protein fold-change coupling $r = +0.020$, $n = 799$). The transcriptional association stops at translation, consistent with buffering.

6.9 [E-E27] Protein-layer null

Condition-dependent quantitative proteomics (Schmidt et al. (2016); 22 conditions, triplicates, $n = 366$): protein-level correlation $r = -0.083$ against transcript $+0.419$ on the same genes. The transcript-level association does not propagate to the protein layer; buffering at translation is the consistent model, positioning Kochanowski et al. (2013) as prior art for the protein-layer null.

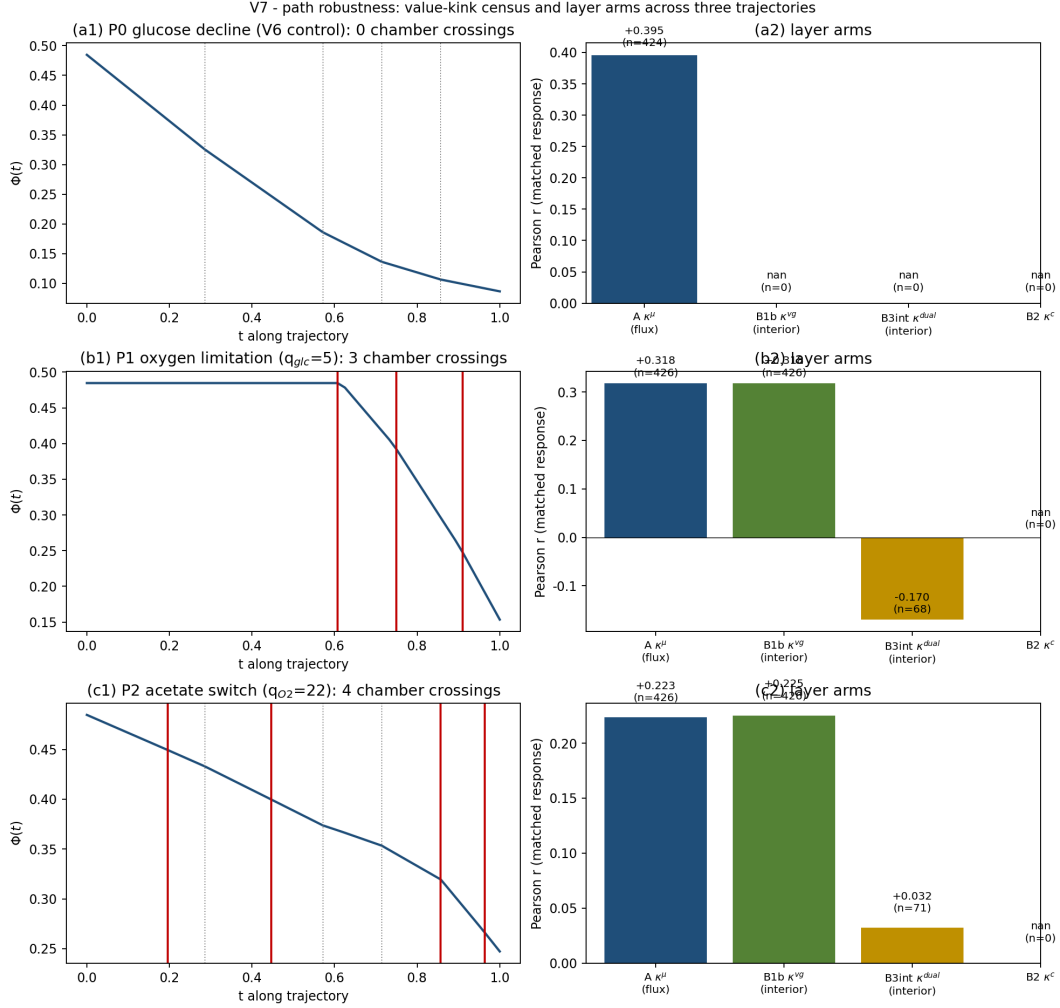


Figure 4: [E-V7] Path robustness. Three trajectories through the same LP (P0 glucose decline, P1 oxygen limitation, P2 acetate switch) with matched M3D responses: the one-chamber property is path-dependent (P1: 3 chamber crossings; P2: 4), while the layer decision and the flux-layer association survive on every path; the P1/P2 predictors correlate with the E24 carbon response at $r = +0.378/+0.391$. Artifact: `deepseek_bridge/v7_path_robustness.png / v7_path_robustness.json`.

7 Discussion: one measure, several resolutions

The three sensitivity objects of the earlier literature — the geometric κ_V , the flux-statistical κ_{flux} , and the time-course κ_{time} — are one measure at different resolutions: the atomic measure of Definition 3.1, its coarse-grainings $\mu * \phi_\sigma$, and its integrals along parameter trajectories (κ^μ). The smooth member exists only through the window $h \ll \sigma \ll L_{\text{var}}$; the $\sigma \rightarrow 0$ limit is the atomic object. Open items, demoted to their honest status: Conjecture 4.6 (general complex stability), E28 under the (ε, σ) design law, and E32 (event-measure stabilization — the L^1 -reconstruction regime of Theorem 4.1(iv)). The formal identity $\kappa_{\text{flux}} = \mathcal{F}[\mu]$ is closed by Theorem 3.11 (the exact objective contraction $D^2\Phi = \sum_r c_r D^2v_r^*$), which supersedes the earlier metric-invariance argument for why the association stands on the flux layer.

Limitations. Correlational design; single-organism scope; operational (not combinatorial) active sets; lexicographic tie-breaking is an engine convention (the value-function carrier is immune, the flux layer declares it — Remark 3.5 — and the association is stable across four alternative selection rules, E-V8); the one-chamber structure of the E24 path is condition-specific (E-V7) while

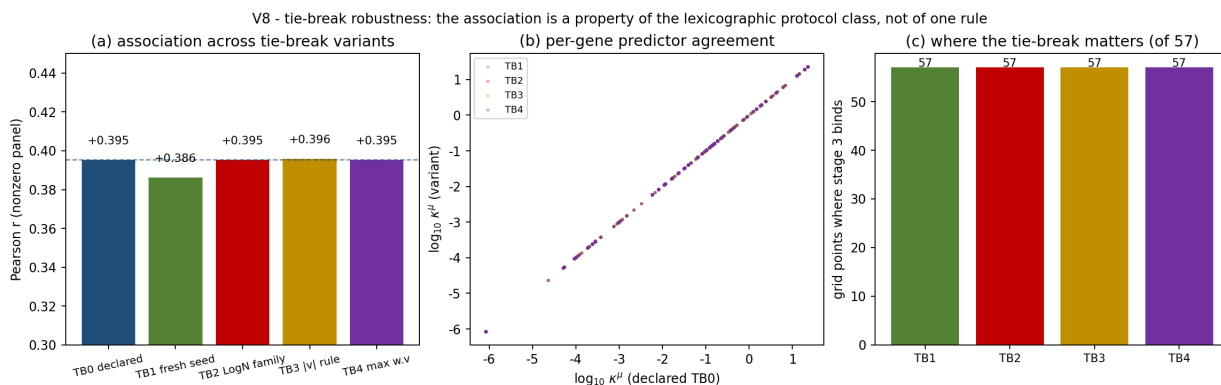


Figure 5: [E-V8] Tie-break robustness. (a) The association across the five stage-3 selection rules (declared plus four variants); (b) per-gene predictor agreement between the declared rule and each variant; (c) the number of grid points at which the tie-break binds (of 57). The value layer is invariant to 0.0 across all variants. Artifact: `deepseek_bridge/v8_tiebreak_robustness.png` / `v8_tiebreak_robustness.json`.

the layer decision itself is path-robust; the cross-platform carbon-switch dissociation (PRECISE, E-V5) bounds the generality of the association; the E22 active-reaction census (438/2,583 and the GPR class counts) is plain-FBA and solver-version-sensitive at the ± 1 -reaction level, while the gene-level panel (435) is stable (numeric audit, Appendix A).

The companion document. The categorical and homotopy-theoretic material of the predecessor manuscript (optic-category composition, the 2-category $\mathbf{StCon}(B)$ of stratified connections, the gluing and composition theorems, the filtered-colimit RAF construction, the $\mathbf{HoTT}/\infty$ -categorical extension) is *preserved, not discarded*: it is extracted, with its status labels (proved / conjecture / machine-verified), into a separate companion paper (*A Categorical Companion to the Measure-Theoretic Curvature Framework*, compiled from the frozen v21 source), which shares no load-bearing content with the present manuscript. The full frozen v21 (10,830 lines) and its complete Git history remain committed as the audit baseline; the split is a publication choice, not a deletion.

What this paper is and is not. The mathematics is classical: parametric linear programming (chamber complexes, active sets, Danskin duality (Danskin, 1967)), Alexandrov’s theorem (Alexandrov, 1939), and distributional second derivatives of piecewise-affine maps; Theorem 3.11 is an elementary identity. The contribution is the systematic *application* of this machinery to flux-balance models (Varma and Palsson, 1994; Orth et al., 2010; Lewis et al., 2010) — the two-layer measure structure of the value/flux pair, the exact coupling identity, the signed extension to arbitrary GPR structure — together with the empirical association and its layer decision. No new measure-theoretic theorem is claimed; the nearest prior art (phenotype phase planes (Edwards et al., 2001; Ibarra et al., 2002), mpLP value-function theory (Borrelli et al., 2003), discrete Monge–Ampère (Gutiérrez, 2001), Regge-type defect measures (Regge, 1961; Cheeger et al., 1984)) is cited where it applies.

8 Methods

8.1 The lexicographic pFBA engine

All deterministic trajectories are computed by a purpose-built three-stage lexicographic solver on the variable split $x = (v, f, r)$ with $v = f - r$, $f, r \geq 0$, stoichiometry $Sv = 0$, and flux bounds

$\ell(\theta) \leq v \leq u(\theta)$ entering affinely through the uptake bounds. The stages are strict lexicographic optimizations, each pinned at the previous stage’s optimum:

$$\begin{aligned} (1) \quad & \mu^* = \max c_{\text{bio}}^\top v && \text{(growth)} \\ (2) \quad & s^2 = \min \sum_r (f_r + r_r) \quad \text{s.t. } v_{\text{bio}} \geq \mu^*(1 - \varepsilon_\mu) && \text{(parsimony)} \\ (3) \quad & \min w^\top v \quad \text{s.t. } \sum_r (f_r + r_r) \leq s^2(1 + \varepsilon_s) && \text{(tie-break)} \end{aligned}$$

with $\varepsilon_\mu = \varepsilon_s = 10^{-9}$ and $w \sim U(0.5, 1.5)$ drawn once with seed 20240901 (the M-series protocol). Stage 3 exists because plain pFBA is vertex-degenerate on these models: repeated warm-started solves of the same LP flip 0.69 reactions on average between vertices of the optimal face. The LPs are solved with HiGHS (Huangfu and Hall, 2018) via `scipy.optimize.linprog`, cold-started and deterministic; a presolve-disabled retry handles the rare infeasibility mis-report. The lex optimum exists, is unique by construction, and is continuous piecewise affine in θ (Lemma 3.10); the tie-break is part of the metric’s definition (Remark 3.5), and its robustness is measured in E-V8. The E22 active-reaction census is the one plain-FBA computation retained (historical protocol of the precursor metric family); it is solver-version-sensitive at the ± 1 -reaction level, while the gene-level panel is stable.

8.2 Models and data provenance

Models. *E. coli* iJO1366 (2,583 reactions) (Orth et al., 2011) for the E-series association experiments; iML1515 (Monk et al., 2017) for the M-series computational batteries (M1, M3, M4) and the V1/Theorem-C verification cut. Genome-scale models are loaded from the local BiGG JSON copies with SHA-256 manifests.

M3D microarray compendium. The E24/E25 expression panels come from the M3D *E. coli* compendium v4 Build 6 (Faith et al., 2008, 2007): 4,297 gene probes $\times$ 907 arrays, $\log_2$ uniformly normalized. The carbon-depletion response is the maximum $|\log_2 \text{FC}|$ over the four stationary time points (135/330/480/720 min) against the MOPS glucose reference; the oxygen-limitation and acetate-switch responses use the M9 WT anaerobic-vs-aerobic and MOPS acetate-vs-glucose contrasts. The reference-level confound control is the mean log-phase WT MOPS glucose expression. The 117 MB compendium tarball is not committed (repository size); it is re-downloadable from the recorded URL with the committed SHA-256 (e73088f7...).

PRECISE. The cross-platform arm uses the ten WT non-glucose carbon conditions of the PRECISE compendium (Sastry et al., 2019) (RNA-seq, TPM). PRECISE has no carbon-depletion class; the dissociation is reported as an honest negative with per-condition values (E-V5).

Proteomics. Condition-dependent quantitative proteomics from Schmidt et al. (2016) (22 conditions, triplicates); integrated protein abundance from PaxDb (Wang et al., 2015); the GSE64021 (Barrett et al., 2013) time course supplies the RNA-seq carbon-starvation and the quantitative proteomics arms (E-E25/E-E26).

8.3 Panel construction and GPR association

The E22 panel is built by reaction-based sampling: solve the lexicographic trajectory on the E22 physiology (glucose 5.0 $\rightarrow$ 1.0, oxygen 22 $\rightarrow$ 5; eight anchors, $8 \times$ per-segment refinement for the V-series), take the 438 reactions active ($|v| > 10^{-9}$ at any grid point) on the plain-FBA E22 census, classify each active reaction’s GPR rule (single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34), and assign each gene the maximum over its associated reactions. The E24 panel is the 433 genes of this set carrying M3D response data; the primary correlation uses the 424 genes with nonzero κ^μ . The near-colliding gene/reaction counts (424/433/435/438/440/525/537) are disambiguated once in the counts ledger of Appendix A.

8.4 Statistical protocols

The predictor is $\log_{10}$ of the per-gene κ^μ (its distribution is log-linear over five decades); the response is $\max |\log_2 \text{FC}|$ of the matched contrast. Reported statistics: Pearson r with analytic p , Spearman ρ_S (raw predictor ranks, full panel and nonzero subset as labeled), partial r controlling the M3D reference level by linear residualization of both variables, permutation p (10^5 label permutations, Monte Carlo), bootstrap 95% CIs (10^4 resamples), and top-vs-bottom decile contrasts (one-sided Mann–Whitney U). All tests are two-sided except the decile contrast. The PRECISE per-condition arms use the same protocol with TPM log-fold changes. Numbers in the manuscript trace to committed artifacts via the numeric audit (`scripts/audit_v2_numbers.py` → `download/deepseek_bridge/v2_number_audit.json`; 73 checks).

8.5 Reproducibility

All experiments deposit artifacts (npz/csv/json/png) with generating scripts under `scripts/` and `download/`; independent-seed reproductions are recorded for the refinement prototype (BT1) and the stress batteries. The V6/V7/V8 layer decisions, the PRECISE arm, and the tie-break robustness re-run in minutes each (`scripts/e24_layer_decision.py`, `v7_path_robustness.py`, `precise_arm_kappamu.py`, `v8_tiebreak_robustness.py`). The full numeric audit (`audit_v2_numbers.py`) re-derives every manuscript number from the committed artifacts (73 checks, 8 repaired defects D-N1–D-N8, one counts ledger) and is re-runnable end-to-end.

A The counts ledger

Near-colliding counts of the manuscript, each traced to a single committed artifact (numeric audit, 2026-09-02):

Value	Meaning (source)
2,583	iJO1366 reactions (model file; recomputed E22-2R)
438	reactions active on the E22 physiology, 17.0% (v15 script printout; plain-FBA, ± 1 solver sensitivity)
440	reactions with $D_2 > 10^{-8}$ along the E24 trajectory (reaction-level event count; V5/V8 json <code>n_events_rxns</code>)
433	panel genes with M3D response (E24 csv row count)
424	panel genes with nonzero κ^μ — the n of the primary correlation (v5/v6/v8 json <code>n_nonzero</code>)
426	nonzero genes on the P1/P2 paths (V7 json)
435	genes with ≥ 1 active reaction, E22 (e22 csv; recomputed exactly)
454	max active reactions across the nine E23 conditions (v16 results json)
537	union of active reactions across conditions, 20.8% (v16 results json; the frozen v21’s 538 is a typo)
525	genes with nonzero κ across conditions (v16 results json; the frozen v21’s 524 is a typo)
1,516	iML1515 genes, M3 single knockouts (m3 json)
2,779	double-knockout pairs, five panels (m3 json)

B Proofs and proof sketches

B.1 Proof of Theorem 3.11

$\Phi = c^\top v^*$ holds pointwise by optimality. Both sides are continuous piecewise affine (Lemma 3.10 for v^* ; LP value function theory for Φ), and differentiation of distributions is linear: applying D^2 to the pointwise identity gives the measure identity $D^2\Phi = \sum_r c_r D^2v_r^*$. Restricting to a crease

facet F gives the per-crease form $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$ at every crease time t_k of any piecewise-affine trajectory, and the contraction bound $|\Delta\Phi'| \leq \|c\|_\infty \|\Delta v'\|_1$ is Hölder’s inequality. (i) An event with unique optima on both sides is a transversal optimal-vertex switch; its slope jump satisfies $c^\top \Delta v' = \Delta\Phi' \neq 0$ unless the value function is affine through it, which would force the optimum to persist — hence objective-invisibility ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) occurs exactly inside ≥ 1 -dimensional optimal faces, where the tie-break selects among vertices without moving the objective; Theorem 3.11(ii) is the sparse-objective special case. The regularity input (continuity and piecewise affinity of v^*) is measured, not assumed: segment residuals $\leq 8 \times 10^{-14}$ (E-M1 glucose) and Φ piecewise affine at 4.2×10^{-13} (E-V1). $\square$

B.2 Proof of Proposition 3.12 (collapse)

If f is λ -semiconvex, then $D^2f + \lambda I \cdot \text{Leb} \succeq 0$ as symmetric-matrix measures. For continuous piecewise-affine f , D^2f is supported on the codimension-one skeleton, hence purely singular with respect to Lebesgue; testing the inequality on skeleton sets forces the singular part itself PSD, so $D^2f \succeq 0$. Mollifying, $D^2(f * \rho_\varepsilon) = (D^2f) * \rho_\varepsilon \succeq 0$, and $f * \rho_\varepsilon \rightarrow f$ uniformly, so f is convex as a uniform limit of convex functions. The converse is trivial. The numerical form: the smallest violating scale $h_{\max}(\lambda)$ of the OR-plus-cap value function satisfies $\lambda h_{\max} = 1/(2\lambda)$ exactly (AX-10), i.e. no finite semiconvexity constant exists at any scale. The signed-layer statement is elementary distribution theory: a continuous PWL function has distributional D^2 supported on its crease skeleton, a finite symmetric Radon measure with per-crease sign flags. $\square$

B.3 Proof of Proposition 3.13

At a generic codimension-two vertex of a convex PWL f on $\mathbb{R}^2$, exactly three facets meet with gradients n_1, n_2, n_3 (cyclically ordered). The Monge–Ampère atom is the volume of the normal fan $\text{conv}\{n_i\}$ (the gradient image sweeps the triangle). Writing $j_i = n_{i+1} - n_i$ for the codimension-one jumps, twice the fan area is $|\det(j_1, j_2)|$ (the shoelace formula on the triangle), giving atom $= \frac{1}{2} |\det(j_1, j_2)| = \frac{1}{2} |j_1| |j_2| \sin \angle$; the product $|j_1| |j_2|$ overestimates by $2/\sin \angle \geq 2$ with equality iff $j_1 \perp j_2$. The simplex-fan generalization to dimension p is the same shoelace argument on a p -simplex fan. Machine verification: 400 random max-of-affine vertices, determinant = fan area to 8.9×10^{-16} , the $2/\sin \angle$ law to 2.5×10^{-12} , orthogonal edge case exactly 2. $\square$

C Layer-1 porting map

v2 section	v21 source (frozen; do not modify)
§1	v21 sec 1 (rewritten to the single story)
§2	v21 sec 2 (subset: active-set notation only)
§3	v21 active-set stratification defs; new D1/D2
§4	new (Theorem B’; replaces the smooth bridge)
§5	M1/M3/M4 report bodies
§6	v21 E22–E27 sections (E22, E23, E25, E26 ported; E24 $\rightarrow$ E-V5 with the PRECISE arm; E27 $\rightarrow$ E-E27; V6/V7/V8 new)
§7	v21 discussion, rewritten as resolution statement
§8	v21 methods; engine documentation (expanded)
Appendices A–B companion	new (numeric audit ledger; proofs of Theorem C and Propositions S, M) categorical/HoTT sections of v21 (preserved in <i>A Categorical Companion</i> , separate document)

References

- Alexander D. Alexandrov. Almost everywhere existence of the second differential of a convex function and some properties of convex functions connected with it. *Leningrad University Annals (Mathematics Series)*, 6:3–35, 1939. In Russian.
- Tanya Barrett, Stephen E. Wilhite, Pierre Ledoux, Carlos Evangelista, Irene F. Kim, Maxim Tomashevsky, Kimberly A. Marshall, Kathy H. Phillippy, Patti M. Sherman, Nicole Holko, et al. NCBI GEO: archive for functional genomics data sets—update. *Nucleic Acids Research*, 41(D1):D991–D995, 2013.
- Patrick Billingsley. *Convergence of Probability Measures*. Wiley, New York, 2nd edition, 1999.
- Francesco Borrelli, Alberto Bemporad, and Manfred Morari. Geometric algorithm for multi-parametric linear programming. *Journal of Optimization Theory and Applications*, 118(3):515–540, 2003.
- Jeff Cheeger, Werner Müller, and Robert Schrader. On the curvature of piecewise flat spaces. *Communications in Mathematical Physics*, 92(3):405–454, 1984.
- John M. Danskin. *The Theory of Max-Min and its Application to Weapons Allocation Problems*. Springer, Berlin, 1967.
- J. S. Edwards, R. U. Ibarra, and B. O. Palsson. In silico predictions of *escherichia coli* metabolic capabilities are consistent with experimental data. *Nature Biotechnology*, 19(2):125–130, 2001.
- Joshua J. Faith, Brian Hayete, Joshua T. Thaden, Itai Mogno, Jamey Wierzbowski, Guillaume Cottarel, Simon Kasif, J. J. Collins, and Timothy S. Gardner. Large-scale mapping and validation of *escherichia coli* transcriptional regulation from a compendium of expression profiles. *PLoS Biology*, 5(1):e8, 2007.
- Joshua J. Faith, Madeline E. Driscoll, Vincent A. Fusaro, Elizabeth J. Cosgrove, Brian Hayete, Frank S. Juhn, and Timothy S. Gardner. Many microarrays database: Affymetrix microarray data and statistical processing routines for transcriptional network analysis. *Nucleic Acids Research*, 36(suppl_1):D866–D870, 2008.
- Cristian E. Gutiérrez. *The Monge-Ampère Equation*. Birkhäuser, Boston, 2001.
- Qijun Huangfu and Julian A. J. Hall. Parallelizing the dual revised simplex method. *Mathematical Programming Computation*, 10(1):119–142, 2018.
- Rafael U. Ibarra, Jeremy S. Edwards, and Bernhard O. Palsson. *Escherichia coli* K-12 undergoes adaptive evolution to achieve in silico predicted optimal growth. *Nature*, 420(6912):186–189, 2002.
- Karl Kochanowski, Benjamin Volkmer, Laura Gerosa, Bob R. Haverkamp van Rijsewijk, Alexander Schmidt, and Matthias Heinemann. Functioning of a metabolic flux sensor in *escherichia coli*. *Proceedings of the National Academy of Sciences*, 110(6):2050–2055, 2013.
- Nathan E. Lewis, Kim K. Hixson, Tom M. Conrad, Joshua A. Lerman, Pep Charusanti, Ashoka D. Polpitiya, Joshua N. Adkins, Gunnar Schramm, Samuel O. Purvine, Daniel Lopez-Ferrer, et al. Omic data from evolved *e. coli* are consistent with computed optimal growth from genome-scale metabolic network models. *Molecular Systems Biology*, 6:390, 2010.
- Jonathan M. Monk, Colton J. Lloyd, Elizabeth Misra, Zachary A. King, Tejas Sandberg, Roger Walker, Daniel Tenenbaum, Karsten Zengler, Ying Hefner, and Bernhard O. Palsson. iML1515, a knowledgebase that computes *escherichia coli* traits. *Nature Biotechnology*, 35(10):904–908, 2017.

- Jeffrey D. Orth, Ines Thiele, and Bernhard O. Palsson. What is flux balance analysis? *Nature Biotechnology*, 28(3):245–248, 2010.
- Jeffrey D. Orth, Tom M. Conrad, Jessica Na, Joshua A. Lerman, Hojung Lofgren, Nathan E. Lewis, Alexander DeLuna, Shuxia A, Yoseph Latif, Brian Liebig, et al. A comprehensive genome-scale reconstruction of *escherichia coli* metabolism—2011. *Molecular Systems Biology*, 7:535, 2011.
- Tullio Regge. General relativity without coordinates. *Il Nuovo Cimento*, 19(3):558–571, 1961.
- R. Tyrrell Rockafellar. *Convex Analysis*. Princeton University Press, Princeton, 1970.
- Aarash V. Sastry, Yichao Gao, Richard Szubin, Ying Hefner, Peng Xu, Huaiyu Dang, and Bernhard O. Palsson. The *escherichia coli* transcriptome, proteome, and metabolome in precise knockouts of every transcription factor (precise 1.0). *bioRxiv*, 2019. 10.1101/080929.
- Alexander Schmidt, Karl Kochanowski, Silke Vedelaar, Erik Ahrné, Benjamin Volkmer, Rémy Calloni, Ioana Chiorescu, Tzong-Yi Cui, Viktor Benes, Hyun-Seok Huh, et al. The quantitative and condition-dependent *escherichia coli* proteome. *Nature Biotechnology*, 34(1):104–110, 2016.
- Daniel Segrè, Dennis Vitkup, and George M. Church. Analysis of optimality in natural and perturbed metabolic networks. *Proceedings of the National Academy of Sciences*, 99(23):15112–15117, 2002.
- Abhishek Varma and Bernhard O. Palsson. Stoichiometric flux balance models quantitatively predict growth and metabolic by-product secretion in wild-type *escherichia coli* W3110. *Applied and Environmental Microbiology*, 60(10):3724–3731, 1994.
- Cédric Villani. *Optimal Transport: Old and New*. Springer, Berlin, 2009.
- Mora Wang, Christian J. Herrmann, Milan Simonovic, Damian Szklarczyk, and Christian von Mering. Version 4.0 of PaxDb: protein abundance data, integrated across model organisms, with a much expanded coverage of the *arabidopsis thaliana* proteome. *Proteomics*, 15(18):3143–3146, 2015.
- Günter M. Ziegler. *Lectures on Polytopes*. Springer, New York, 1995.